ETSI TS 103 920-2 V1.2.1 (2026-07)



TECHNICAL SPECIFICATION

Core Network and Interoperability Testing (INT);
5G NGAP Conformance Testing for the N2 interface;
(3GPP™ Release 16);
Part 2: Test Suite Structure (TSS) and Test Purposes (TP)

Reference
RTS/INT-00218

Keywords
conformance, NGAP, TSS&TP

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

Contents

Intellectual Property Rights 5
Foreword 5
Modal verbs terminology 5
1 Scope 6
2 References 6
2.1 Normative references 6
2.2 Informative references 6
3 Definition of terms, symbols and abbreviations 7
3.1 Terms 7
3.2 Symbols 7
3.3 Abbreviations 7
4 Test configurations 7
4.1 Introduction 7
4.2 Test configuration using the N2 interface 7
4.3 Network infrastructure 9
5 Test Suite Structure (TSS) and Test Purposes (TP) 10
5.1 Test Suite Structure 10
5.1.1 TP naming convention 10
5.1.2 Test strategy 10
5.1.3 TP structure 10
5.2 Test Purposes 11
5.2.0 Introduction 11
5.2.1 PICS references 12
5.2.2 N2 interface 12
5.2.2.1 gNB Role 12
5.2.2.1.1 Test selection 12
5.2.2.1.2 Events for gNB Role 12
5.2.2.1.3 PDU Session Management Procedures 12
5.2.2.1.4 UE Context Management Procedures 13
5.2.2.1.5 UE Mobility Management Procedures 14
5.2.2.1.6 Paging Procedures 14
5.2.2.1.7 Transport of NAS Messages Procedures 14
5.2.2.1.8 Interface Management Procedures 14
5.2.2.1.9 Configuration Transfer Procedure 15
5.2.2.1.10 Warning Message Transmission Procedures 16
5.2.2.1.11 NRPPa Transport Procedures 16
5.2.2.1.12 Trace Procedures 16
5.2.2.1.13 Location Reporting Procedures 16
5.2.2.1.14 UE TNLA Binding Procedures 16
5.2.2.1.15 UE Radio Capability Management Procedures 16
5.2.2.1.16 Data Usage Reporting Procedures 16
5.2.2.1.17 RIM Information Transfer Procedures 16
5.2.2.2 AMF Role 17
5.2.2.2.1 Test selection 17
5.2.2.2.2 Events for AMF Role 17
5.2.2.2.3 PDU Session Management Procedures 17
5.2.2.2.4 UE Context Management Procedures 22
5.2.2.2.5 UE Mobility Management 24
5.2.2.2.6 Paging Procedures 24
5.2.2.2.7 Transport of NAS Messages Procedures 25
5.2.2.2.8 Interface Management Procedures 25
5.2.2.2.9 Configuration Transfer Procedure 31
5.2.2.2.10 Warning Message Transmission Procedures 31
5.2.2.2.11 NRPPa Transport 31

5.2.2.2.12 Trace Procedures 31
5.2.2.2.13 Location Reporting 31
5.2.2.2.14 UE TNLA Binding Procedures 31
5.2.2.2.15 UE Radio Capability Management 31
5.2.2.2.16 Data Usage Reporting Procedures 31
5.2.2.2.17 RIM Information Transfer Procedures 31

Annex A (normative): Test Purpose source files 32

History 33

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Core Network and Interoperability Testing (INT).

The present document is part 2 of a multi-part deliverable. Full details of the entire series can be found in part 1 [2].

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

1 Scope

The present document provides the Test Suite Structure (TSS) and Test Purposes (TP) for the test specification for the NGAP protocol on the N2 interface as specified in ETSI TS 138 413 [1] in compliance with the relevant requirements and in accordance with the relevant guidance given in ISO/IEC 9646-7 [i.2] and ETSI ETS 300 406 [i.3].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 138 413 (V16.17.0): "5G; NG-RAN; NG Application Protocol (NGAP) (3GPP TS 38.413 version 16.17.0 Release 16)".

[2] ETSI TS 103 920-1: "Core Network and Interoperability Testing (INT); 5G NGAP Conformance Testing for the N2 interface; (3GPPTM Release 16); Part 1: Protocol Implementation Conformance Statement (PICS)".

[3] ETSI TS 123 502 (V16.19.0): "5G; Procedures for the 5G System (5GS) (3GPP TS 23.502 version 16.19.0 Release 16)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ISO/IEC 9646-1: "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 1: General concepts".

[i.2] ISO/IEC 9646-7: "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 7: Implementation Conformance Statements".

[i.3] ETSI ETS 300 406: "Methods for testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TS 138 413 [1] and the following apply:

Abstract Test Method (ATM): Refer to ISO/IEC 9646-1 [i.1].

Abstract Test Suite (ATS): Refer to ISO/IEC 9646-1 [i.1].

Implementation Under Test (IUT): Refer to ISO/IEC 9646-1 [i.1].

Test Purpose (TP): Refer to ISO/IEC 9646-1 [i.1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TS 138 413 [1] and the following apply:

TS	Test System
TSS	Test Suite Structure

4 Test configurations

4.1 Introduction

Test purposes of the present document address the 5G functional entities gNB and AMF that are accessible via the standardized N2 interface.

4.2 Test configuration using the N2 interface

The N2 interface is located between the gNB and the AMF. Following configurations are simplified to highlight tested interface and involved entities. Overall network infrastructure is shown under clause 4.3.

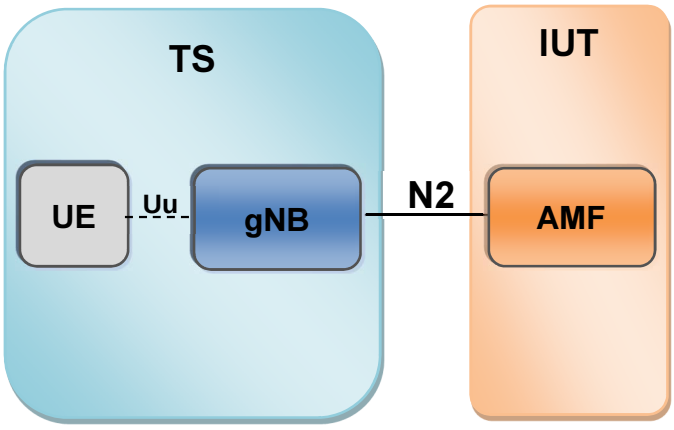


Figure 1: Test configuration CF_AMF_N2

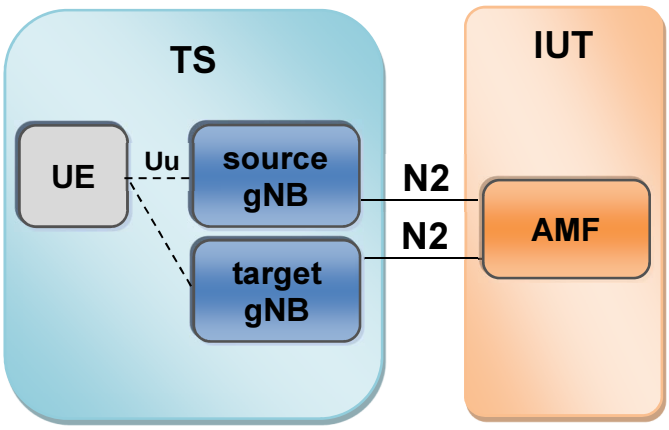


Figure 2: Test configuration CF_AMF_2N2

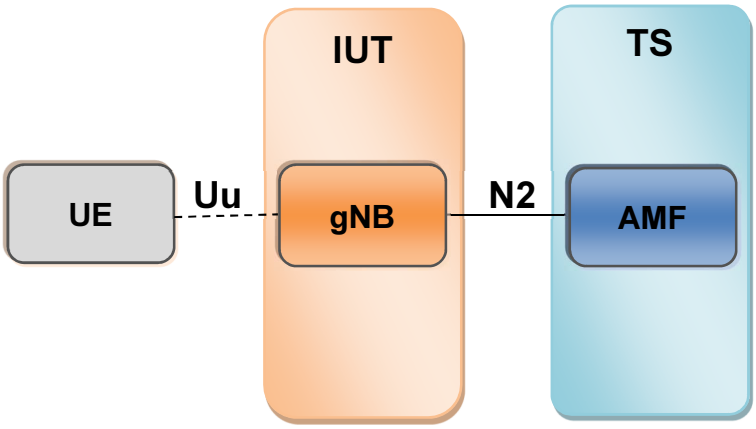


Figure 3: Test configuration CF_GNB_N2

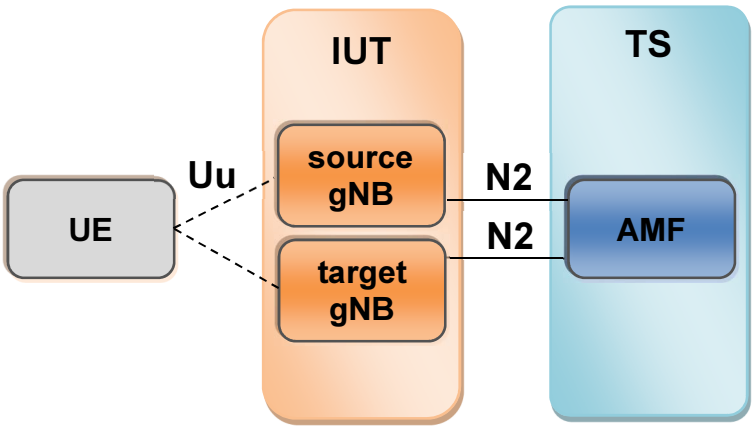


Figure 4: Test configuration CF_GNB_2N2

4.3 Network infrastructure

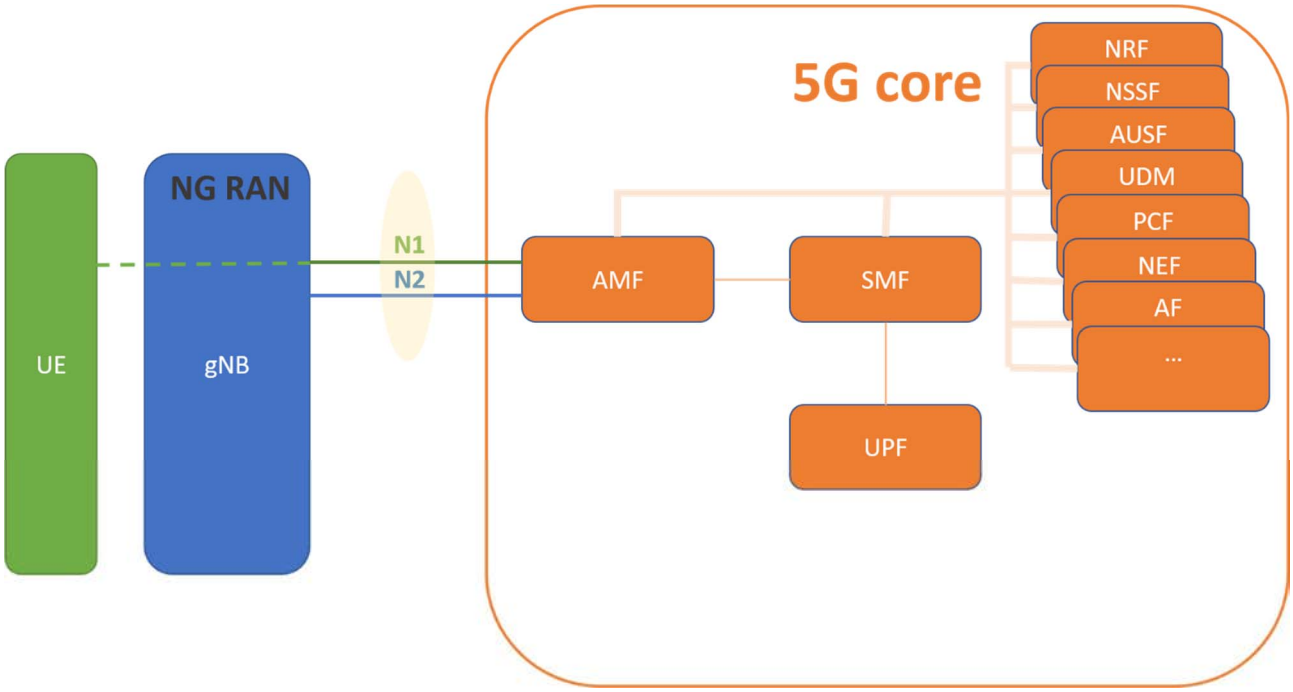


Figure 5: Network architecture

5 Test Suite Structure (TSS) and Test Purposes (TP)

5.1 Test Suite Structure

5.1.1 TP naming convention

TPs are numbered, starting at 01, within each group. Groups are organized according to the TSS.

Table 1: TP identifier naming convention scheme

Identifier: **<TP>_<iut>_<scope>_<nn>[_<n>]**			
<tp> = Test Purpose:	fixed to "TP"		
<interface or protocol>	Interface or protocol: NGAP		
<iut> = type of IUT:	GNB or AMF		
<scope> = group	PDU	PDU Session Management Procedures	
	CMP	Context Management procedures	
	MMP	UE Mobility Management Procedures	
	PAG	Paging	
	NAS	Transport of NAS Messages Procedures	
	IMP	Interface Management Procedures	
	CTP	Configuration Transfer Procedures	
	WTP	Warning Message Transmission Procedures	
	NTP	NRPPa transport Procedures	
	TRP	Trace Procedures	
	LRP	Location Reporting Procedures	
	UBP	UE TNLA Binding Procedures	
	URP	UE Radio Capability Management Procedures	
	DRP	Data Usage Reporting Procedures	
	RIP	RIM Information Transfer Procedures	
<nn> = sequential number	(01 to 99)		
<n> = sequential number variant	(1 to 9)		

5.1.2 Test strategy

As the base specification in ETSI TS 138 413 [1] contains no explicit requirements for testing, the TPs were generated as a result of an analysis of the base standard and the PICS specification ETSI TS 103 920-1 [2].

5.1.3 TP structure

Each TP has been written in a manner which is consistent with all other TPs. The intention of this is to make the TPs more readable and checkable. A particular structure has been used which is illustrated in Table 2. Table 2 should be read in conjunction with any TP, i.e. use a TP as an example to facilitate the full comprehension of Table 2.

Table 2: Structure of a single TP

TP part	Text	Example
Header	**<Identifier>** **<clause number in base ETSI TS 138 413 [1]>** **<PICS reference>**	**see Table 1** **clause 8.3.11.2** **A.3/3.1**
Summary	*Short free text description of the test objective*	Verify that the IUT can send a HANDOVER REQUIRED message containing mandatory IEs due to UE mobility management procedure.
Configuration	*Test configuration as described in clause 4.2*	CFG_GNB_01
Initial condition (optional)	*Free text description of the condition that the IUT has reached before the test purpose applies.*	
Start point	Ensure that the IUT in the <state> *see ETSI TS 138 413 [1], clause 8.1* *and/or further actions before stimulus* *if the action is sending/receiving* *see below for message structure*	Handover Preparation having sent a HANDOVER_REQUIRED
Stimulus	<**trigger**>, *see below for message structure* *or* <goal>	on receipt of a HANDOVER_COMMAND (see note 2)
Reaction	<**action**>. *if the action is sending* *see below for message structure* <next action>, *etc.*	sends, saves, does, etc.
Message structure	<message type> *a)* containing a(n) <IE name> IE (see note 4) *b)* indicating <coding of the field> and *back to a) or b) (see note 3)*	Message exchange, etc. (see note 2)

NOTE 1: Text in italics will not appear in TPs and text between <> is filled in for each TP and may differ from one TP to the next.
NOTE 2: All messages are considered as "valid and compatible" unless otherwise specified in the test purpose. This includes the presence of all mandatory IEs as specified in ETSI TS 138 413 [1].
NOTE 3: An IE can be embedded into another IE. This is expressed by indentations, e.g. if Message1 contains IE1 and IE2 where IE1 has IE3 embedded this will be expressed like this:

```
sends/receives Message 1
    containing IE1
        containing IE3
            indicating ...
    containing IE2
        indicating ...
```

NOTE 4: IE value fields used for e.g. identification or address should be equal in the scope of TP if not stated otherwise.

5.2 Test Purposes

5.2.0 Introduction

The present document contains modified and new Test Purposes. The Test Purpose Identifiers are not backwards compatible with previous versions of the document.

5.2.1 PICS references

All PICS items referred to in this clause are as specified in ETSI TS 103 920-1 [2] unless indicated otherwise by another numbered reference. PICS items are only meant for test selection, therefore only PICS items with status optional or conditional are explicitly mentioned.

5.2.2 N2 interface

5.2.2.1 gNB Role

5.2.2.1.1 Test selection

The IUT takes the role of the gNB; PICS A.2/1.

5.2.2.1.2 Events for gNB Role

Events used in the Test Purposes for the NGAP gNB role are described in Table 3.

Table 3: Events defined for gNB Role

Event	Description
hasEstablished_NG_C_AssociationToAMF	An NG-C (NGAP) association exists between the gNB and AMF, reached after successful NG Setup procedure.
hasEstablishedPDUsession	The AMF maintains an established PDU Session Resource in the UE NGAP context for the concerned PDU Session ID following a successful PDU Session Resource Setup procedure.
hasSuccessfullyEstablishedTNLconnection	A Transport Network Layer (TNL) connection is successfully established after completion of the SCTP association setup between the gNB and the AMF.
havingSentPDUSessionEstablishment	The UE has sent a PDU Session Establishment Request and is awaiting the network response.
isRegisteredToAMF	The UE has completed the 5G Registration procedure and received a Registration Accept from the AMF.
u_Plane_inactivity_timer_expires	The user-plane inactivity timer expires when no uplink or downlink user-plane traffic is detected for a configured duration.

5.2.2.1.3 PDU Session Management Procedures

TP Id	TP_NGAP_GNB_PDU_01
Test Objective	Verify that the GNB successfully processes a PDU_SESSION_RESOURCE_SETUP_REQUEST and answers with PDU_SESSION_RESOURCE_SETUP_RESPONSE for successfully established PDU session.
Reference	ETSI TS 138 413 [1], Clauses 8.2.1.2, 9.2.1.1 and 9.2.1.2
Configuration	CF_GNB_N2
PICS Selection	PICS_A2_1 and PICS_A3_1_1
Initial Conditions	

```
with {
    the UE isRegisteredToAMF and
    the UE havingSentPDUSessionEstablishment
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a PDU_SESSION_RESOURCE_SETUP_REQUEST containing {
            AMF_UE_NGAP_ID set to PX_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to RV_RAN_UE_NGAP_ID,
            PDU_Session_Resource_Setup_Request_List containing {
                PDU_Session_Resource_Setup_Request_Item containing {
                    PDU_SessionId set to PX_PDU_SessionId,
                    PDU_Session_NAS_PDU containing {
                        uL_NAS_TRANSPORT_message_identity set to UL_NAS_TRANSPORT,
                        payload_container containing {
                            Message_type set to PDU_SESSION_ESTABLISHMENT_ACCEPT}},
                    S_NSSAI containing {
```

```
                        SST set to PX_SST,
                        SD set to PX_SD},
                    PDU_Session_Resource_Setup_Request_Transfer containing {
                        UL_NG_U_UP_TNL_Information containing {
                            EndpointIPAddress set to PX_N3_ENDPOINT_IP_ADDRESS,
                            GTP_TEID set to PX_GTP_TEID},
                        PDU_Session_Type set to "IPv4",
                        QoS_Flow_Setup_Request_List containing {
                            QoS_Flow_Setup_Request_Item containing {
                                QoSFlowIdentifier set to PX_QFI,
                                QoSFlowLevelQosParameters containing {
                                    QoSCharactersistics containing {
                                        Non_Dynamic_5QI set to PX_5QI},
                                    Allocation_and_Retention_Priority}}}}}}}
        from the AMF entity
    }
    then {
        the IUT sends a PDU_SESSION_RESOURCE_SETUP_RESPONSE containing {
            AMF_UE_NGAP_ID set to PX_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to RV_RAN_UE_NGAP_ID,
            PDU_Session_Resource_Setup_Response_List containing {
              PDU_Session_Resource_Setup_Response_Item containing {
                PDU_SessionId set to PX_PDU_SessionId,
                PDU_Session_Resource_Setup_Response_Transfer containing {
                  DL_QoS_Flow_per_TNL_Information containing {
                    UP_Transport_Layer_Information containing {
                        EndpointIPAddress set to RV_N3_GNB_ADDRESS,
                        GTP_TEID set to RV_GTP_TEID},
                    Associated_QoS_Flow_List containing {
                        Associated_QoS_Flow_Item containing {
                            QoS_Flow_Identifier set to PX_QFI}}}}}}}
         to the AMF entity
       }
   }
```

5.2.2.1.4 UE Context Management Procedures

TP Id	TP_NGAP_GNB_CMP_07
Test Objective	Verify that the GNB successfully requests the AMF to release the UE-associated logical NG-connection due to user inactivity.
Reference	ETSI TS 138 413 [1], Clauses 8.3.2.2 and 9.2.2.4
Configuration	CF_GNB_N2
PICS Selection	PICS_A2_1 and PICS_A3_2_2
Initial Conditions	

```
with {
    the UE isRegisteredToAMF and
    the UE hasEstablishedPDUsession
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT u_Plane_inactivity_timer_expires
    }
    then {
        the IUT sends a UE_CONTEXT_RELEASE_REQUEST containing {
            AMF_UE_NGAP_ID set to PX_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to RV_RAN_UE_NGAP_ID,
            PDU_Session_Resource_List containing {
                PDU_Session_Resource_Item containing {
                    PDU_SessionId set to PX_PDU_SessionId}},
            Cause set to "RadioNetwork.user-inactivity"}
        to the AMF entity
    }
}
```

TP Id	TP_NGAP_GNB_CMP_07_2
Test Objective	Verify that the GNB successfully requests the AMF to release the UE-associated logical NG-connection after rejected registration procedure.
Reference	ETSI TS 138 413 [1], Clauses 8.3.2.2 and 9.2.2.4 ETSI TS 123 502 [3], Clause 4.2.6
Configuration	CF_GNB_N2
PICS Selection	PICS_A2_1 and PICS_A3_2_2
Initial Conditions	

```
with {
    the GNB hasEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the AMF sends a DOWNLINK_NAS_TRANSPORT containing {
            NAS_PDU containing {
                Message_identity set to REGISTRATION_REJECT}}
    }
    then {
        the IUT sends a UE_CONTEXT_RELEASE_REQUEST containing {
            AMF_UE_NGAP_ID set to PX_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to RV_RAN_UE_NGAP_ID,
            PDU_Session_Resource_List containing {
                PDU_Session_Resource_Item containing {
                    PDU_SessionId set to PX_PDU_SessionId}},
            Cause set to "RadioNetwork.unspecified"}
        to the AMF entity
    }
}
```

5.2.2.1.5 UE Mobility Management Procedures

Void.

5.2.2.1.6 Paging Procedures

Void.

5.2.2.1.7 Transport of NAS Messages Procedures

Void.

5.2.2.1.8 Interface Management Procedures

TP Id	TP_NGAP_GNB_IMP_01
Test Objective	Verify that the GNB sends an NG_SETUP_REQUEST message to the AMF after TNL establishment.
Reference	ETSI TS 138 413 [1], Clauses 8.7.1.2 and 9.2.6.1
Configuration	CF_GNB_N2
PICS Selection	PICS_A2_1 and PICS_A3_6_1
Expected Behaviour	

```
ensure that {
    when {
        the IUT hasSuccessfullyEstablishedTNLconnection
    }
    then {
        the IUT sends an NG_SETUP_REQUEST containing {
            Global_RAN_Node_ID containing {
                    Global_gNB_ID containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        gNB_ID set to PX_gNB_ID}},
            Supported_TA_List containing {
                Supported_TA_Item containing {
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
                            PLMN_Identity set to PX_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
```

```
                                    SST set to PX_SST,
                                    SD set to PX_SD
                                    OR NOT SD}}}}}},
           Default_Paging_DRX set to PX_GNB_DEF_DRX}
        to the AMF entity
    }
}
```

TP Id	TP_NGAP_GNB_IMP_05
Test Objective	Verify that the GNB successfully processes a full NG_RESET message and answers with NG_RESET_ACKNOWLEDGE to acknowledge the reset.
Reference	ETSI TS 138 413 [1], Clauses 8.7.4.2.1, 9.2.6.11 and 9.2.6.12
Configuration	CF_GNB_N2
PICS Selection	PICS_A2_1 and PICS_A3_6_4

Initial Conditions

```
with {
    the GNB hasEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_RESET containing {
            Cause set to PX_CAUSE,
            CHOICE_Reset_Type containing {
                NG_interface containing {
                    Reset_All set to "Reset all"}}}
    from the AMF entity
    }
    then {
        the IUT sends an NG_RESET_ACKNOWLEDGE containing {
            not UE_associated_Logical_NGconnection_List}
        to the AMF entity
    }
}
```

TP Id	TP_NGAP_GNB_IMP_05_2
Test Objective	Verify that the GNB successfully processes a partial NG_RESET message and answers with NG_RESET_ACKNOWLEDGE to acknowledge the partial reset.
Reference	ETSI TS 138 413 [1], Clauses 8.7.4.2.1, 9.2.6.11 and 9.2.6.12
Configuration	CF_GNB_N2
PICS Selection	PICS_A2_1 and PICS_A3_6_4

Initial Conditions

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_RESET containing {
            Cause set to PX_CAUSE,
            CHOICE_Reset_Type containing {
                Part_of_NG_interface containing {
                    UE_associated_Logical_NGconnection_List containing {
                        UE_associated_Logical_NGconnection_Item containing {
                            AMF_UE_NGAP_ID set to PX_AMF_UE_NGAP_ID,
                            RAN_UE_NGAP_ID set to RV_RAN_UE_NGAP_ID}}}}}
    from the AMF entity
    }
    then {
        the IUT sends an NG_RESET_ACKNOWLEDGE containing {
            not UE_associated_Logical_NGconnection_List}
        to the AMF entity
    }
}
```

5.2.2.1.9 Configuration Transfer Procedure

Void.

5.2.2.1.10 Warning Message Transmission Procedures

Void.

5.2.2.1.11 NRPPa Transport Procedures

Void.

5.2.2.1.12 Trace Procedures

Void.

5.2.2.1.13 Location Reporting Procedures

Void.

5.2.2.1.14 UE TNLA Binding Procedures

Void.

5.2.2.1.15 UE Radio Capability Management Procedures

TP Id	TP_NGAP_GNB_URP_01
Test Objective	Verify that the IUT sends correct UE_RADIO_CAPABILITY_INFO_INDICATION.
Reference	ETSI TS 138 413 [1], Clauses 8.14.1.2 and 9.2.13.1
Configuration	CF_GNB_N2
PICS Selection	PICS_A2_1 and PICS_A3_13_1
Initial Conditions	

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a INITIAL_CONTEXT_SETUP_REQUEST containing {
            not UE_Radio_Capability}
    }
    then {
        the IUT sends a UE_RADIO_CAPABILITY_INFO_INDICATION containing {
            AMF_UE_NGAP_ID set to PX_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to RV_RAN_UE_NGAP_ID,
            UE_Radio_Capability containing {
                UERadioAccessCapabilityInformation containing {
                    criticalExtensions containing {
                        c1 containing {
                            ueRadioAccessCapabilityInformation containing {
                                UE_CapabilityRAT_ContainerList containing {
                                    UE_CapabilityRAT_Container containing {
                                      rat_Type set to "nr",
                                      UE_NR_Capability containing {
                                          accessStratumRelease set to "rel16"}}}}}}}}}
        to the AMF entity
       }
   }
```

5.2.2.1.16 Data Usage Reporting Procedures

Void.

5.2.2.1.17 RIM Information Transfer Procedures

Void.

5.2.2.2 AMF Role

5.2.2.2.1 Test selection

The IUT takes the role of the AMF; PICS A.2/2.

5.2.2.2.2 Events for AMF Role

Events used in the Test Purposes for the NGAP AMF role are described in Table 4.

Table 4: Events defined for AMF Role

Event	Description
hasActiveUeNgapContext	The AMF maintains a valid UE NGAP context identified by the AMF UE NGAP ID and RAN UE NGAP ID and associated with an active UE-associated logical NG-connection towards the NG-RAN node.
hasEstablished_NG_C_AssociationToAMF	An NG-C (NGAP) association exists between the gNB and AMF, reached after successful NG Setup procedure.
hasEstablishedInitialContext	The UE context is established in the gNB after successful Initial Context Setup procedure triggered by the AMF.
hasEstablishedPduSessionResourceForTheConcernedPduSessionId	The AMF maintains an established PDU Session Resource in the UE NGAP context for the concerned PDU Session ID following a successful PDU Session Resource Setup procedure.
hasEstablishedTNL_AssociationToAMF	A transport-layer (SCTP) association between gNB and AMF is established before any NGAP procedures.
hasNoPduSessionResourceEstablishedForTheConcernedPduSessionId	The AMF does not maintain any established PDU Session Resource in the UE NGAP context for the concerned PDU Session ID, and no NGAP PDU Session Resource Setup procedure is ongoing for that PDU Session ID.
hasNotEstablished_NG_C_AssociationToAMF	No NG-C (NGAP) association exists between the gNB and AMF.
hasSentPDU_SESSION_RESOURCE_RELEASE_COMMAND	The AMF has transmitted a PDU_SESSION_RESOURCE_RELEASE_COMMAND message to the NG-RAN node for the concerned UE and is awaiting the corresponding PDU_SESSION_RESOURCE_RELEASE_RESPONSE.
hasSentPDU_SESSION_RESOURCE_SETUP_REQUEST	The AMF has transmitted a PDU_SESSION_RESOURCE_SETUP_REQUEST message to the NG-RAN node for the concerned UE and is awaiting the corresponding PDU_SESSION_RESOURCE_SETUP_RESPONSE.
initiatesPduSessionResourceModificationToAddQoSFlow	AMF initiates PDU_SESSION_RESOURCE_MODIFY_REQUEST to add the QoS flow for already established PDU session.
isInCM_CONNECTED	UE is in CM_CONNECTED state.
isNotRegisteredToAMF	The UE has no active 5G registration with the AMF.
isRegisteredToAMF	The UE has completed the 5G Registration procedure and received a Registration Accept from the AMF.
processTheUENasDeregistration	UE performs the NAS Deregistration signalling to release its registration and associated contexts from the AMF.

5.2.2.2.3 PDU Session Management Procedures

TP Id	TP_NGAP_AMF_PDU_01
Test Objective	Verify that the IUT successfully processes a PDU Session Establishment Request and initiates the PDU Session Resource Setup procedure for a single PDU_Session_Resource.
Reference	ETSI TS 138 413 [1], Clauses 8.2.1.2 and 9.2.1.1
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_1_1
Initial Conditions	

with {
the UE isInCM_CONNECTED and
the IUT hasActiveUeNgapContext and
the IUT hasNoPduSessionResourceEstablishedForTheConcernedPduSessionId
}	

Expected Behaviour	
ensure that { when {	

```
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            NAS_PDU containing {
                ul_nas_transport_message_identity set to UL_NAS_TRANSPORT,
                payload_container containing {
                    PDU_session_identity set to PX_PDU_SessionId,
                    Message_type set to PDU_SESSION_ESTABLISHMENT_REQUEST}},
            User_Location_Information containing {
                nr_cgi containing {
                    PLMN_Identity set to PX_PLMN_Identity},
                tai containing {
                    PLMN_Identity set to PX_PLMN_Identity,
                    TAC set to PX_TAC}
            }
        }
        from the UE entity
    }
    then {
        the IUT sends a PDU_SESSION_RESOURCE_SETUP_REQUEST containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            PDU_Session_Resource_Setup_Request_List containing {
                PDU_Session_Resource_Setup_Request_Item containing {
                    PDU_SessionId set to PX_PDU_SessionId,
                    PDU_Session_NAS_PDU containing {
                        uL_NAS_TRANSPORT_message_identity set to UL_NAS_TRANSPORT,
                        payload_container containing {
                            PDU_session_identity set to PX_PDU_SessionId,
                            Message_type set to PDU_SESSION_ESTABLISHMENT_ACCEPT,
                            S_NSSAI containing {
                               SST set to PX_SST,
                               SD set to PX_SD}}},
                    S_NSSAI containing {
                        SST set to PX_SST,
                        SD set to PX_SD},
                    PDU_Session_Resource_Setup_Request_Transfer containing {
                        PDU_Session_Type set to IPv4,
                        QoS_Flow_Setup_Request_List containing {
                            QoS_Flow_Setup_Request_Item containing {
                                QoSFlowIdentifier set to PX_QFI,
                                QoSFlowLevelQosParameters containing {
                                    QoSCharacteristics containing {
                                        Non_Dynamic_5QI set to PX_5QI},
                                     AllocationAndRetentionPriority}}}}}}}
         to the GNB entity
     }
 }
```

TP Id	TP_NGAP_AMF_PDU_01_2
Test Objective	Verify that the IUT successfully receives and processes the PDU_SESSION_RESOURCE_SETUP_RESPONSE message and completes the PDU Session Resource Setup procedure for a single PDU_Session_Resource.
Reference	ETSI TS 138 413 [1], Clauses 8.2.1.2 and 9.2.1.1
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_1_1
Initial Conditions	

```
with {
    the UE isInCM_CONNECTED and
    the IUT hasActiveUeNgapContext and
    the IUT hasSentPDU_SESSION_RESOURCE_SETUP_REQUEST
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT sends a PDU_SESSION_RESOURCE_SETUP_REQUEST containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            PDU_Session_Resource_Setup_Request_List containing {
                PDU_Session_Resource_Setup_Request_Item containing {
                    PDU_SessionId set to PX_PDU_SessionId,
                    PDU_Session_NAS_PDU containing {
                        uL_NAS_TRANSPORT_message_identity set to UL_NAS_TRANSPORT,
                        payload_container containing {
                            Message_type set to PDU_SESSION_ESTABLISHMENT_ACCEPT}},
```

```
                    S_NSSAI containing {
                        SST set to PX_SST,
                        SD set to PX_SD},
                    PDU_Session_Resource_Setup_Request_Transfer containing {
                        PDU_Session_Type set to IPv4,
                        QoS_Flow_Setup_Request_List containing {
                            QoS_Flow_Setup_Request_Item containing {
                                QoSFlowIdentifier set to RV_QFI,
                                QoSFlowLevelQosParameters containing {
                                    QoSCharacteristics containing {
                                        Non_Dynamic_5QI set to PX_5QI},
                                     AllocationAndRetentionPriority}}}}}}}
        to the GNB entity
    }
    then {
        the IUT accepts a PDU_SESSION_RESOURCE_SETUP_RESPONSE containing {
           AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
           RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
           PDU_Session_Resource_Setup_Response_List containing {
               PDU_Session_Resource_Setup_Response_Item containing {
                   PDU_SessionId set to PX_PDU_SessionId,
                   PDU_Session_Resource_Setup_Response_Transfer containing {
                       DL_QoS_Flow_per_TNL_Information containing {
                           AssociatedQoSFlowList containing {
                               AssociatedQoSFlowItem containing {
                                   QoSFlowIdentifier set to RV_QFI}}}}}}}
        from the GNB entity
     }
}
```

TP Id	TP_NGAP_AMF_PDU_02
Test Objective	Verify that the IUT successfully processes a PDU Session Release request and initiates the PDU Session Resource Release procedure for a single PDU_Session_Resource.
Reference	ETSI TS 138 413 [1], Clauses 8.2.2.2 and 9.2.1.3
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_1_2

Initial Conditions

```
with {
    the UE isInCM_CONNECTED and
    the IUT hasActiveUeNgapContext and
    the IUT hasEstablishedPduSessionResourceForTheConcernedPduSessionId
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an UPLINK_NAS_TRANSPORT containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            NAS_PDU containing {
                ul_nas_transport_message_identity set to UL_NAS_TRANSPORT,
                payload_container containing {
                    PDU_SessionId set to PX_PDU_SessionId,
                    Message_type set to PDU_SESSION_RELEASE_REQUEST}},
            User_Location_Information containing {
                nr_cgi containing {
                    PLMN_Identity set to PX_PLMN_Identity},
                tai containing {
                    PLMN_Identity set to PX_PLMN_Identity,
                    TAC set to PX_TAC}
            }
        }
        from the GNB entity
    }
    then {
        the IUT sends a PDU_SESSION_RESOURCE_RELEASE_COMMAND containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            NAS_PDU containing {
                ul_nas_transport_message_identity set to DL_NAS_TRANSPORT,
                payload_container containing {
                    Message_type set to PDU_SESSION_RELEASE_COMMAND},
                PDU_SessionId set to PX_PDU_SessionId},
            PDU_Session_Resource_To_Release_List containing {
              PDU_Session_Resource_To_Release_Item containing {
                PDU_SessionId set to PX_PDU_SessionId,
```

```
                PDU_Session_Resource_Release_Command_Transfer containing {
                  Cause set to PX_Cause}}}} //normal_release
         to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_PDU_02_2
Test Objective	Verify that the IUT successfully receives and processes the PDU_SESSION_RESOURCE_RELEASE_RESPONSE message and completes the PDU Session Resource Release procedure for a single PDU_Session_Resource.
Reference	ETSI TS 138 413 [1], Clauses 8.2.2.2 and 9.2.1.3
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_1_2

Initial Conditions

```
with {
    the UE isInCM_CONNECTED and
    the IUT hasActiveUeNgapContext and
    the IUT hasSentPDU_SESSION_RESOURCE_RELEASE_COMMAND
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT sends a PDU_SESSION_RESOURCE_RELEASE_COMMAND containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            NAS_PDU containing {
                ul_nas_transport_message_identity set to DL_NAS_TRANSPORT,
                payload_container containing {
                    Message_type set to PDU_SESSION_RELEASE_COMMAND},
                PDU_SessionId set to PX_PDU_SessionId},
            PDU_Session_Resource_To_Release_List containing {
              PDU_Session_Resource_To_Release_Item containing {
                PDU_SessionId set to PX_PDU_SessionId,
                PDU_Session_Resource_Release_Command_Transfer containing {
                  Cause set to PX_Cause}}}} //normal_release
         to the GNB entity
    }
    then {
        the IUT accepts a PDU_SESSION_RESOURCE_RELEASE_RESPONSE containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            PDU_Session_Resource_To_Release_List containing {
              PDU_Session_Resource_To_Release_Item containing {
                PDU_SessionId set to PX_PDU_SessionId}}}
         from the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_PDU_03
Test Objective	Verify that the IUT initiates the PDU Session Resource Modify procedure to request the setup of an additional QoS flow for an already established PDU session. NOTE: The UE shall be configured in the IUT to require an additional QoS flow for the concerned PDU session.
Reference	ETSI TS 138 413 [1], Clauses 8.2.3.2 and 9.2.1.5
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_1_3

Initial Conditions

```
with {
    the UE isInCM_CONNECTED and
    the IUT hasActiveUeNgapContext and
    the IUT hasEstablishedPduSessionResourceForTheConcernedPduSessionId
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT initiatesPduSessionResourceModificationToAddQoSFlow
    }
    then {
        the IUT sends a PDU_SESSION_RESOURCE_MODIFY_REQUEST containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            PDU_Session_Resource_Modify_Request_List containing {
```

```
                PDU_Session_Resource_Modify_Request_Item containing {
                    PDU_SessionId set to PX_PDU_SessionId,
                    PDU_Session_Resource_Modify_Request_Transfer containing {
                        QoSFlowAddOrModifyRequestList containing {
                            QoSFlowAddOrModifyRequestItem containing {
                                QoSFlowIdentifier set to RV_2nd_QFI,
                                QoSFlowLevelQosParameters containing {
                                    QoSCharacteristics containing {
                                        Non_Dynamic_5QI set to PX_2nd_5QI,
                                    AllocationAndRetentionPriority}}}}}}}}
        to the GNB entity
        and the IUT accepts a PDU_SESSION_RESOURCE_MODIFY_RESPONSE containing {
           AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
           RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
           PDU_Session_Resource_Modify_Response_List containing {
               PDU_Session_Resource_Modify_Response_Item containing {
                   PDU_SessionId set to PX_PDU_SessionId,
                   PDU_Session_Resource_Modify_Response_Transfer containing {
                       QoSFlowAddOrModifyResponseList containing {
                           QoSFlowAddOrModifyResponseItem containing {
                               QoSFlowIdentifier set to RV_2nd_QFI}}}}}}
        from the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_PDU_04
Test Objective	Verify that the AMF node successfully processes a PDU_SESSION_RESOURCE_MODIFY_INDICATION and answers with PDU_SESSION_RESOURCE_MODIFY_CONFIRM for successfully modified PDU session.
Reference	ETSI TS 138 413 [1], Clauses 8.2.5.2 and 9.2.1.8
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_1_5

Initial Conditions

```
with {
    the UE isInCM_CONNECTED and
    the IUT hasActiveUeNgapContext and
    the IUT hasEstablishedPduSessionResourceForTheConcernedPduSessionId
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a PDU_SESSION_RESOURCE_MODIFY_INDICATION containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            PDU_Session_Resource_Modify_Indication_List containing {
              PDU_Session_Resource_Modify_Indication_Item containing {
                PDU_SessionId set to PX_PDU_SessionId,
                PDU_Session_Resource_Modify_Indication_Transfer containing {
                  DLQoSFlowperTNLInformation containing {
                    UPTransportLayerInformation containing {
                        EndpointIPAddress set to PX_N3_GNB_IP_ADDRESS,
                        GTP_TEID set to RV_GTP_TEID},
                    AssociatedQoSFlowList containing {
                      AssociatedQoSFlowItem containing {
                        QoSFlowIdentifier set to RV_QFI}}}}}}
        from the GNB entity
    }
    then {
        the IUT sends a PDU_SESSION_RESOURCE_MODIFY_CONFIRM containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            PDU_Session_Resource_Modify_Confirm_List containing {
              PDU_Session_Resource_Modify_Confirm_Item containing {
                PDU_SessionId set to PX_PDU_SessionId,
                PDU_Session_Resource_Modify_Confirm_Transfer containing {
                  QoSFlowModifyConfirmList containing {
                    QoSFlowModifyConfirmItem containing {
                      QoSFlowIdentifier set to RV_QFI}},
                  UL_NG_U_UP_TNLInformation containing {
                    UP_Transport_Layer_Information containing {
                        EndpointIPAddress set to PX_N3_AMF_IP_ADDRESS,
                        GTP_TEID set to RV_GTP_TEID}}}}}
        to the GNB entity
      }
   }
```

TP Id	TP_NGAP_AMF_PDU_05
Test Objective	Verify that the AMF node processes a PDU_SESSION_RESOURCE_MODIFY_INDICATION and answers with PDU_SESSION_RESOURCE_MODIFY_CONFIRM for not successfully modified PDU session.
Reference	ETSI TS 138 413 [1], Clauses 8.2.5.2 and 9.2.1.8
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_1_5
Initial Conditions	

with {
the UE isInCM_CONNECTED and
the IUT hasActiveUeNgapContext and
the IUT hasEstablishedPduSessionResourceForTheConcernedPduSessionId
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a PDU_SESSION_RESOURCE_MODIFY_INDICATION containing {
AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
PDU_Session_Resource_Modify_Indication_List containing {
PDU_Session_Resource_Modify_Indication_Item containing {
PDU_SessionId set to PX_PDU_SessionId,
PDU_Session_Resource_Modify_Indication_Transfer containing {
DLQoSFlowperTNLInformation containing {
UPTransportLayerInformation containing {
EndpointIPAddress set to PX_N3_GNB_IP_ADDRESS,
GTP_TEID set to RV_GTP_TEID},
AssociatedQoSFlowList containing {
AssociatedQoSFlowItem containing {
QoSFlowIdentifier set to PX_QFI_Wrong}}}}}}}
from the GNB entity
}
then {
the IUT sends a PDU_SESSION_RESOURCE_MODIFY_CONFIRM containing {
AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
PDU_Session_Resource_Failed_To_Modify_List containing {
PDU_Session_Resource_Failed_To_Modify_Item containing {
PDU_SessionId set to PX_PDU_SessionId,
PDU_Session_Resource_Modify_Indication_Unsuccessful_Transfer containing {
Cause set to "Unknown QoS Flow ID"}}}}
to the GNB entity
}
}	

5.2.2.2.4 UE Context Management Procedures

TP Id	TP_NGAP_AMF_CMP_01
Test Objective	Verify the IUT correctly sets the UE Context after Authentication and Security procedure has been executed.
Reference	ETSI TS 138 413 [1], Clause 8.3.1.2
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_2_1
Initial Conditions	

with {
the GNB hasEstablished_NG_C_AssociationToAMF and
the IUT hasActiveUeNgapContext and
the UE isNotRegisteredToAMF
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an UplinkNASTransport containing {
AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
NAS_PDU containing {
Message_identity set to SECURITY_MODE_COMPLETE},
User_Location_Information containing {
nr_cgi containing {
PLMN_Identity set to PX_PLMN_Identity},
tai containing {	

```
                    PLMN_Identity set to PX_PLMN_Identity,
                    TAC set to PX_TAC}}}
          from the GNB entity
    }
    then {
        the IUT sends an INITIAL_CONTEXT_SETUP_REQUEST containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            GUAMI containing {
                PLMN_Identity set to PX_PLMN_Identity,
                AMF_Region_ID set to PX_AMF_Region_ID,
                AMF_Set_ID set to PX_AMF_Set_ID,
                AMF_Pointer set to PX_AMF_Pointer_ID},
            Allowed_NSSAI containing {
                Allowed_NSSAI_List containing {
                    Allowed_NSSAI_Item containing {
                        S_NSSAI containing {
                            SST set to PX_SST_Allowed,
                            SD set to PX_SD_Allowed}}}},
            UE_Security_Capabilities containing {
                NR_Encryption_Algorithms set to PX_NR_Encryption_Algorithms,
                NR_Integrity_Protection_Algorithms set to PX_NR_Integrity_Protection_Algorithms,
                E_UTRA_Encryption_Algorithms set to PX_E_UTRA_Encryption_Algorithms,
                E_UTRA_Integrity_Protection_Algorithms set to
PX_E_UTRA_Integrity_Protection_Algorithms},
            Security_Key set to RV_Security_Key,
            NAS_PDU containing {
                Message_identity set to REGISTRATION_ACCEPT}}
        to the GNB entity
        and the IUT accepts a INITIAL_CONTEXT_SETUP_RESPONSE containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID}
        from the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_CMP_02
Test Objective	Verify that the IUT can send a UE_CONTEXT_RELEASE_COMMAND that contains both the AMF_UE_NGAP_ID IE and the RAN_UE_NGAP_ID IE or only AMF_UE_NGAP_ID .
Reference	ETSI TS 138 413 [1], Clauses 8.3.3 and 9.2.2.5
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_2_3

Initial Conditions

```
with {
    the UE isRegisteredToAMF and
    the IUT hasActiveUeNgapContext and
    the UE hasEstablishedInitialContext
}
```

Expected Behaviour

```
ensure that {
    when {
        // implicit trigger: AMF releases UE Context
        the IUT processTheUENasDeregistration
    }
    then {
        the IUT sends a UE_CONTEXT_RELEASE_COMMAND containing {
            CHOICE_UE_NGAP_IDs containing {
                UE_NGAP_ID_pair containing {
                    AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
                    RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID
                }
            },
            Cause set to "nas.normal_release"
        }
        to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_CMP_03
Test Objective	Verify that the IUT executes a UE_CONTEXT_RELEASE procedure when it has received UE_CONTEXT_RELEASE_REQUEST due to GNB generated reasons.
Reference	ETSI TS 138 413 [1], Clauses 8.3.2, 8.3.3 and 9.2.2.5
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_2_3

Initial Conditions

```
with {
    the UE isRegisteredToAMF and
    the UE hasEstablishedInitialContext
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a UE_CONTEXT_RELEASE_REQUEST containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            PDU_Session_Resource_List containing {
                PDU_Session_Resource_Item containing {
                    PDU_SessionId set to PX_PDU_SessionId}},
            Cause set to PX_VA_Cause}
        from the GNB entity
    }
    then {
        the IUT sends a UE_CONTEXT_RELEASE_COMMAND containing {
            CHOICE_UE_NGAP_IDs containing {
                UE_NGAP_ID_pair containing {
                    AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
                    RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID}}
            ,
            Cause set to PX_VA_Cause}
        to the GNB entity
        and the IUT accepts a UE_CONTEXT_RELEASE_COMPLETE containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID}
    }
}
```

Table 5: Variants for TP_NGAP_AMF_CMP_03

TP Id	PX_VA_Cause
TP_NGAP_AMF_CMP_03_ID_01	"RadioNetwork.user-inactivity"
TP_NGAP_AMF_CMP_03_ID_02	"RadioNetwork.unspecified"
TP_NGAP_AMF_CMP_03_ID_03	"RadioNetwork.release due to NG-RAN generated reason"
TP_NGAP_AMF_CMP_03_ID_04	"RadioNetwork.cell not available"
TP_NGAP_AMF_CMP_03_ID_05	"RadioNetwork.radio resource not available"
TP_NGAP_AMF_CMP_03_ID_06	"RadioNetwork.radio connection with UE lost"

5.2.2.2.5 UE Mobility Management

Void.

5.2.2.2.6 Paging Procedures

Void.

5.2.2.2.7 Transport of NAS Messages Procedures

TP Id	TP_NGAP_AMF_NAS_01
Test Objective	Verify that the IUT can send a DOWNLINK_NAS_TRANSPORT message to carry NAS information over the NG interface after INITIAL_UE_MESSAGE to start with UE registration procedure.
Reference	ETSI TS 138 413 [1], Clauses 8.6.2.2, 9.2.5.1 and 9.2.5.2 ETSI TS 123 502 [3], Clause 4.24.1
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_5_2

Initial Conditions

```
with {
    the GNB hasEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an INITIAL_UE_MESSAGE containing {
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            NAS_PDU containing {
                Message_identity set to REGISTRATION_REQUEST},
            User_Location_Information containing {
                nr_cgi containing {
                    PLMN_Identity set to PX_PLMN_Identity},
                tai containing {
                    PLMN_Identity set to PX_PLMN_Identity,
                    TAC set to PX_TAC}},
            RRC_Establishmnet_Cause set to "mo-Signalling"}
        from the GNB entity
    }
    then {
        the IUT sends a DOWNLINK_NAS_TRANSPORT containing {
            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID,
            NAS_PDU containing {
                Message_identity set to AUTHENTICATION_REQUEST}}
        to the GNB entity
    }
}
```

5.2.2.2.8 Interface Management Procedures

TP Id	TP_NGAP_AMF_IMP_01
Test Objective	Verify that the AMF node successfully processes an NG_SETUP_REQUEST message containing the PLMN identity, TAC, and supported slice IEs, and responds with an NG_SETUP_RESPONSE message to acknowledge the setup.
Reference	ETSI TS 138 413 [1], Clauses 8.7.1.2, 9.2.6.1 and 9.2.6.2
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_1

Initial Conditions

```
with {
    the GNB hasNotEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_SETUP_REQUEST containing {
            Global_RAN_Node_ID containing {
                    Global_gNB_ID containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        gNB_ID set to PX_gNB_ID}},
            Supported_TA_List containing {
                Supported_TA_Item containing {
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
                            PLMN_Identity set to PX_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST,
                                    SD set to PX_SD}}}}}
            }}
```

```
        from the GNB entity
    }
    then {
        the IUT sends an NG_SETUP_RESPONSE containing {
            AMF_Name set to PX_AMF_Name,
            Served_GUAMI_List containing {
                Served_GUAMI_Item containing {
                    GUAMI containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        AMF_Region_ID set to PX_AMF_Region_ID,
                        AMF_Set_ID set to PX_AMF_Set_ID,
                        AMF_Pointer set to PX_AMF_Pointer_ID}}},
            PLMN_Support_List containing {
                PLMN_Support_Item containing {
                    PLMN_Identity set to PX_PLMN_Identity,
                    Slice_Support_List containing {
                        S_NSSAI containing {
                                SST set to PX_SST,
                                SD set to PX_SD}}}}}
         to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_IMP_01_2
Test Objective	Verify that the AMF node successfully processes an NG_SETUP_REQUEST message containing the PLMN identity, TAC, supported slice IEs and Extended RAN Node Name, and responds with an NG_SETUP_RESPONSE message to acknowledge the setup.
Reference	ETSI TS 138 413 [1], Clauses 8.7.1.2, 9.2.6.1 and 9.2.6.2
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_1

Initial Conditions

```
with {
    the GNB hasNotEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_SETUP_REQUEST containing {
            Global_RAN_Node_ID containing {
                    Global_gNB_ID containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        gNB_ID set to PX_gNB_ID}},
            RAN_Node_Name set to PX_RAN_Node_Name,
            Supported_TA_List containing {
                Supported_TA_Item containing {
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
                            PLMN_Identity set to PX_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST,
                                    SD set to PX_SD}}}}},
            Extended_RAN_Node_Name set to PX_Extended_RAN_Node_Name
            }}
        from the GNB entity
    }
    then {
        the IUT sends an NG_SETUP_RESPONSE containing {
            AMF_Name set to PX_AMF_Name,
            Served_GUAMI_List containing {
                Served_GUAMI_Item containing {
                    GUAMI containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        AMF_Region_ID set to PX_AMF_Region_ID,
                        AMF_Set_ID set to PX_AMF_Set_ID,
                        AMF_Pointer set to PX_AMF_Pointer_ID}}},
            PLMN_Support_List containing {
                PLMN_Support_Item containing {
                    PLMN_Identity set to PX_PLMN_Identity,
                    Slice_Support_List containing {
                        S_NSSAI containing {
                                SST set to PX_SST,
                                SD set to PX_SD}}}}}
         to the GNB entity
```

```
    }
}
```

TP Id	TP_NGAP_AMF_IMP_01_3
Test Objective	Verify that the AMF with IAB support successfully processes an NG_SETUP_REQUEST message containing the PLMN identity, TAC and supported slice IEs, and responds with an NG_SETUP_RESPONSE message containing IAB Supported IE to acknowledge the setup.
Reference	ETSI TS 138 413 [1], Clauses 8.7.1.2, 9.2.6.1 and 9.2.6.2
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_1 and PICS_A4_6_1_2

Initial Conditions

```
with {
    the GNB hasNotEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_SETUP_REQUEST containing {
            Global_RAN_Node_ID containing {
                    Global_gNB_ID containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        gNB_ID set to PX_gNB_ID}},
            RAN_Node_Name set to PX_RAN_Node_Name,
            Supported_TA_List containing {
                Supported_TA_Item containing {
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
                            PLMN_Identity set to PX_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST,
                                    SD set to PX_SD}}}}}
            }}
        from the GNB entity
    }
    then {
        the IUT sends an NG_SETUP_RESPONSE containing {
            AMF_Name set to PX_AMF_Name,
            Served_GUAMI_List containing {
                Served_GUAMI_Item containing {
                    GUAMI containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        AMF_Region_ID set to PX_AMF_Region_ID,
                        AMF_Set_ID set to PX_AMF_Set_ID,
                        AMF_Pointer set to PX_AMF_Pointer_ID}}},
            PLMN_Support_List containing {
                PLMN_Support_Item containing {
                    PLMN_Identity set to PX_PLMN_Identity,
                    Slice_Support_List containing {
                        S_NSSAI containing {
                            SST set to PX_SST,
                            SD set to PX_SD}}}},
            IAB_Supported set to "true"}
         to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_IMP_02
Test Objective	Verify that the AMF node successfully decline an NG_SETUP_REQUEST message containing only PLMNs/SNPNs that does not recognize or support, rejects the NG Setup procedure by responding with NG_SETUP_FAILURE message including the appropriate cause value.
Reference	ETSI TS 138 413 [1], Clauses 8.7.1.3, 9.2.6.1 and 9.2.6.3
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_1

Initial Conditions

```
with {
    the GNB hasEstablishedTNL_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
```

```
        the IUT receives an NG_SETUP_REQUEST containing {
            Supported_TA_List containing {
                Supported_TA_Item containing {
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
                            PLMN_Identity set to PX_NOTSUPPORTED_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST,
                                    SD set to PX_SD}}}}}
            }}
        from the GNB entity
    }
    then {
        the IUT sends an NG_SETUP_FAILURE containing {
              Cause set to "unknown-PLMN-or-SNPN"}
         to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_IMP_04
Test Objective	Verify that the AMF node successfully processes an NG_SETUP_REQUEST message including the appropriate data and answers with NG_SETUP_RESPONSE to acknowledge the setup.
Reference	ETSI TS 138 413 [1], Clauses 8.7.1.2, 9.2.6.1 and 9.2.6.2
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_1

Initial Conditions

```
with {
    the GNB hasEstablishedTNL_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_SETUP_REQUEST containing {
            Global_RAN_Node_ID containing {
                    Global_gNB_ID containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        gNB_ID set to PX_gNB_ID}},
            RAN_Node_Name set to PX_RAN_NODE_NAME,
            Supported_TA_List containing {
                Supported_TA_Item containing {
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
                            PLMN_Identity set to PX_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST,
                                    SD set to PX_SD}},
                            NPN_Support containing {
                                SNPN containing {
                                    NID set to PX_NID}},
                            Extended_TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST_EXTENDED,
                                    SD set to PX_SD_EXTENDED}}}},
                RAT_Information set to "unlicensed"},
            Extended_RAN_Node_Name containing {
                RAN_Node_Name_Visible set to PX_RAN_NODE_NAME_VISIBLE}}}
        from the GNB entity
    }
    then {
        the IUT sends an NG_SETUP_RESPONSE containing {
            AMF_Name set to PX_AMF_Name,
            Served_GUAMI_List containing {
                Served_GUAMI_Item containing {
                    GUAMI containing {
                        PLMN_Identity set to PX_PLMN_Identity,
                        AMF_Region_ID set to PX_AMF_Region_ID,
                        AMF_Set_ID set to PX_AMF_Set_ID,
                        AMF_Pointer set to PX_AMF_Pointer_ID}}},
            Relative_AMF_Capacity,
            PLMN_Support_List containing {
                PLMN_Support_Item containing {
```

```
                    PLMN_Identity set to PX_PLMN_Identity,
                    Slice_Support_List containing {
                        S_NSSAI containing {
                            SST set to PX_SST,
                            SD set to PX_SD}},
                    NPN_Support containing {
                        SNPN containing {
                            NID set to PX_NID}},
                    Extended_Slice_Support_List  containing {
                        S_NSSAI containing {
                            SST set to PX_SST_EXTENDED,
                            SD set to PX_SD_EXTENDED}}}},
            Extended_RAN_Node_Name containing {
                RAN_Node_Name_Visible set to PX_RAN_NODE_NAME_VISIBLE}}
         to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_IMP_05
Test Objective	Verify that the AMF node successfully processes a RAN_CONFIGURATION_UPDATE message containing new slice with RAN_CONFIGURATION_UPDATE_ACKNOWLEDGE to acknowledge the update with supported slice at AMF.
Reference	ETSI TS 138 413 [1], Clauses 8.7.2.2, 9.2.6.4 and 9.2.6.5
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_2
Initial Conditions	

```
with {
    the GNB hasEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a RAN_CONFIGURATION_UPDATE containing {
            Supported_TA_List containing {
                Supported_TA_Item containing {
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
                            PLMN_Identity set to PX_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST_NEW,
                                    SD set to PX_SD_NEW}}}}}}}
        from the GNB entity
    }
    then {
        the IUT sends a RAN_CONFIGURATION_UPDATE_ACKNOWLEDGE
        to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_IMP_06
Test Objective	Verify that the AMF node successfully declines a RAN_CONFIGURATION_UPDATE message with RAN_CONFIGURATION_UPDATE_FAILURE and appropriate cause value (i.e. slice not supported) when the AMF is not able to handle new slice type as requested.
Reference	ETSI TS 138 413 [1], Clauses 8.7.2.3 and 9.2.6.4 and 9.2.6.6
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_2
Initial Conditions	

```
with {
    the GNB hasEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a RAN_CONFIGURATION_UPDATE containing {
            Supported_TA_List containing {
                Supported_TA_Item containing { // Implementation note: RAN has been reconfigured
via O&M interface to support different SST
                    TAC set to PX_TAC,
                    Broadcast_PLMN_List containing {
                        Broadcast_PLMN_Item containing {
```

```
                            PLMN_Identity set to PX_PLMN_Identity,
                            TAI_Slice_Support_List containing {
                                S_NSSAI containing {
                                    SST set to PX_SST_UNSUPPORTED, // Implementation note: The new
SST indicated is not supported by Core side.
                                    SD set to PX_SD_NEW}}}}}}}
        from the GNB entity
    }
    then {
        the IUT sends a RAN_CONFIGURATION_UPDATE_FAILURE containing {
            Cause set to "RadioNetwork_slice-not-supported"}
        to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_IMP_08
Test Objective	Verify that the AMF node successfully processes a full NG_RESET message and answers with NG_RESET_ACKNOWLEDGE to acknowledge the reset.
Reference	ETSI TS 138 413 [1], Clauses 8.7.4.2.2, 9.2.6.11 and 9.2.6.12
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_4
Initial Conditions	

```
with {
    the GNB hasEstablished_NG_C_AssociationToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_RESET containing {
            Cause set to PX_CAUSE,
            CHOICE_Reset_Type containing {
                NG_interface containing {
                    Reset_All set to "Reset all"}}}
       from the GNB entity
    }
    then {
        the IUT sends an NG_RESET_ACKNOWLEDGE containing {
            not UE_associated_Logical_NGconnection_List}
        to the GNB entity
    }
}
```

TP Id	TP_NGAP_AMF_IMP_08_2
Test Objective	Verify that the AMF node successfully processes a partial NG_RESET message and answers with NG_RESET_ACKNOWLEDGE to acknowledge the partial reset.
Reference	ETSI TS 138 413 [1], Clauses 8.7.4.2.2, 9.2.6.11 and 9.2.6.12
Configuration	CF_AMF_N2
PICS Selection	PICS_A2_2 and PICS_A4_6_4
Initial Conditions	

```
with {
    the UE isRegisteredToAMF
}
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives an NG_RESET containing {
            Cause set to PX_CAUSE,
            CHOICE_Reset_Type containing {
                Part_of_NG_interface containing {
                    UE_associated_Logical_NGconnection_List containing {
                        UE_associated_Logical_NGconnection_Item containing {
                            AMF_UE_NGAP_ID set to RV_AMF_UE_NGAP_ID,
                            RAN_UE_NGAP_ID set to PX_RAN_UE_NGAP_ID}}}}}
        from the GNB entity
    }
    then {
        the IUT sends an NG_RESET_ACKNOWLEDGE containing {
          not UE_associated_Logical_NGconnection_List}
        to the GNB entity
    }
}
```

5.2.2.2.9 Configuration Transfer Procedure

Void.

5.2.2.2.10 Warning Message Transmission Procedures

Void.

5.2.2.2.11 NRPPa Transport

Void.

5.2.2.2.12 Trace Procedures

Void.

5.2.2.2.13 Location Reporting

Void.

5.2.2.2.14 UE TNLA Binding Procedures

Void.

5.2.2.2.15 UE Radio Capability Management

Void.

5.2.2.2.16 Data Usage Reporting Procedures

Void.

5.2.2.2.17 RIM Information Transfer Procedures

Void.

Annex A (normative): Test Purpose source files

All TPs from Clause 5 are defined in the files TP_AMF_NGAP.tdltx and TP_GNB_NGAP.tdltx in a machine-readable format. These files are available at the ETSI Forge repository:

https://forge.etsi.org/rep/int/5g-core/nas/-/tree/v1.2.1/test_purposes?ref_type=tags.

History

Version	Date	Status
V1.1.1	October 2024	Publication
V1.2.1	July 2026	Publication